www.PapaCambridge.com

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

0882405589

ADDITIONAL MATHEMATICS **4037/21**

Paper 2 **May/June 2011**

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
The use of an electronic calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 80.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
Total	

This document consists of **14** printed pages and **2** blank pages.

© UCLES 2011

www.PapaCambridge.com

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

www.PapaCambridge.com

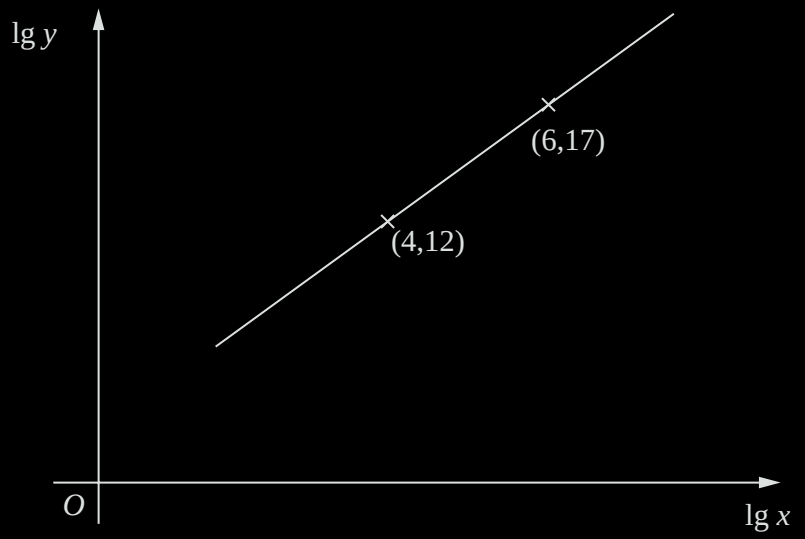
www.PapaCambridge.com

www.PapaCambridge.com

www.PapaCambridge.com

www.PapaCambridge.com

www.PapaCambridge.com



(i) Express y in terms of x, giving your answer in the form $y = ax^b$. [6]

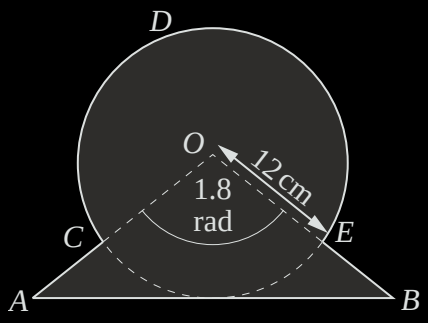
(ii) Find the value of x when $y = 300$. [2]

www.PapaCambridge.com

www.PapaCambridge.com

www.PapaCambridge.com

www.PapaCambridge.com



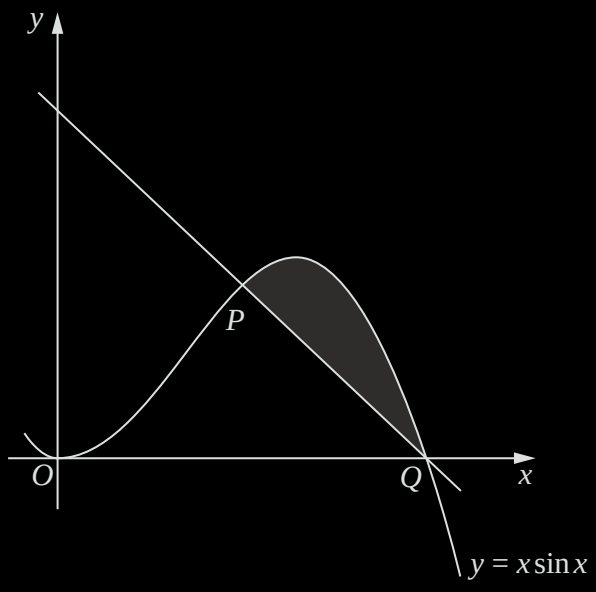
The diagram shows an isosceles triangle AOB and a sector $OCDEO$ of a circle with centre O. The line AB is a tangent to the circle. Angle $AOB = 1.8$ radians and the radius of the circle is 12 cm.

(i) Show that the distance $AC = 7.3$ cm to 1 decimal place. [2]

(ii) Find the perimeter of the shaded region. [6]

(iii) Find the area of the shaded region. [4]

OR



The diagram shows part of the curve $y = x\sin x$ and the normal to the curve at the point $P\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$. The curve passes through the point $Q(\pi, 0)$.

(i) Show that the normal to the curve at P passes through the point Q. [4]

(ii) Given that $\frac{\mathrm{d}}{\mathrm{d}x}(x\cos x) = \cos x - x\sin x$, find $\int x\sin x \mathrm{d}x$. [3]

(iii) Find the area of the shaded region. [5]

www.PapaCambridge.com

www.PapaCambridge.com

www.PapaCambridge.com

www.PapaCambridge.com